



ITI Cmos Analog IC Design
Lab 08
Negative Feedback

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Section 7 online

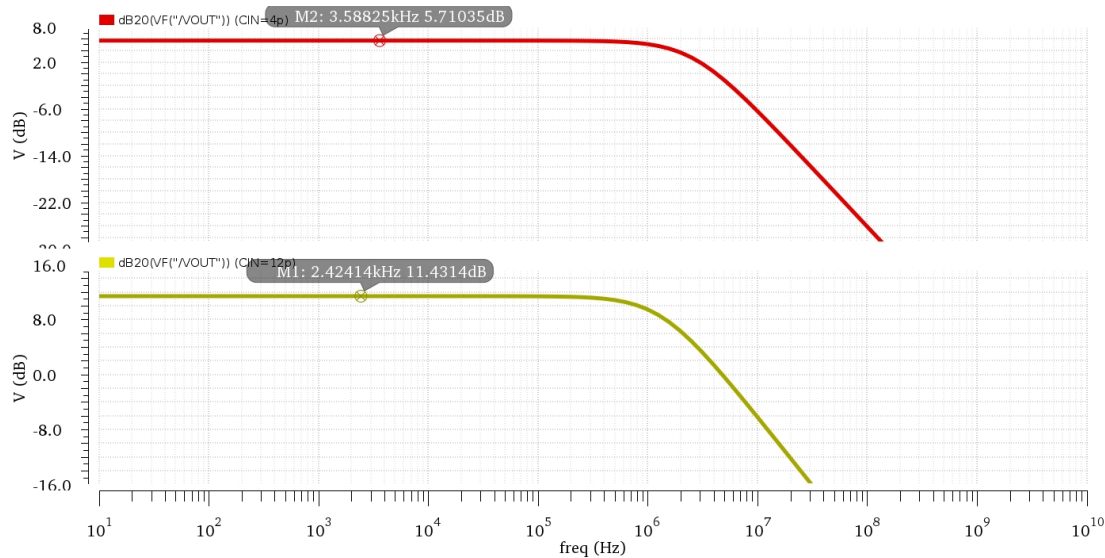
Simulation Parameters

Values used in the AC, STB, and transient testbenches for both Part 1 (behavioral OTA) and Part 2 (real 5T OTA), unless otherwise noted.

Description	Variable	Value
Feedback capacitance	CF	4 pF
Input capacitance	CIN	4 pF, 12 pF
Load capacitance	CL	5 pF
Transconductance of behavioral model	GM	2.86 mS
DC gain of behavioral model	Av	55
UGF of behavioral model	wu	as in Lab 07
Output cap of behavioral model	COUT	GM / wu
Output res of behavioral model	ROUT	Av / GM
AC stimulus magnitude	VINAC	1
Transient stimulus peak	VP	50 mV
Transient stimulus frequency	FIN	1 kHz
Bias current	IB	as in Lab 07
CM input voltage	VICM	middle of the CMIR

Part 1: Feedback with Behavioral OTA

1) Closed-Loop Gain vs. Frequency



- hand analysis

$$Acl = \frac{Aol}{1 + \beta Aol}, \quad \beta = \frac{CF}{CF + CIN}$$

$$BW = wp (1 + \beta Aol), \quad wp = \frac{1}{Rout Cout}$$

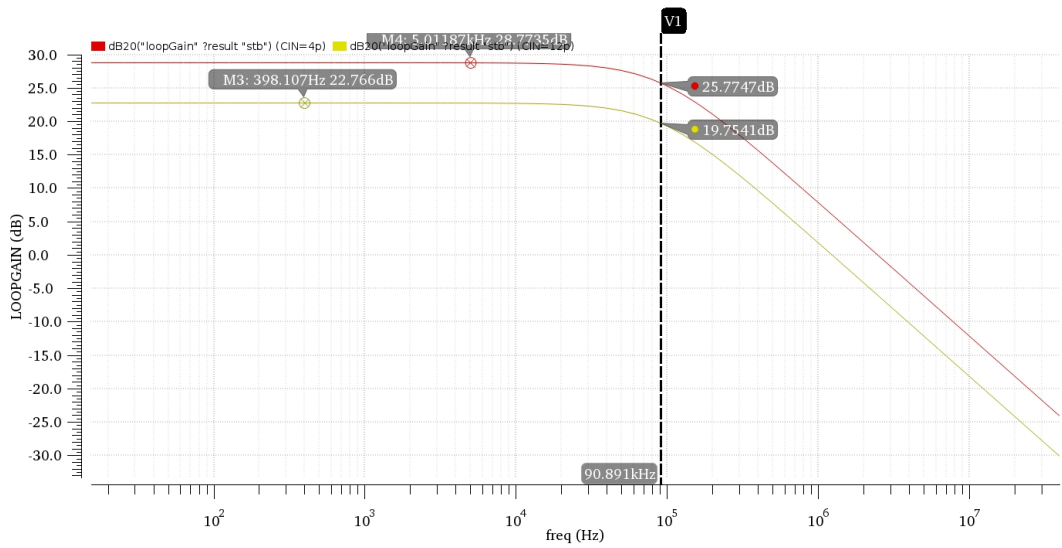
$$GBW = Acl * BW,$$

Parameters: CIN=4p			
1	my_labs:ota_tb:1	/VOUT	
1	my_labs:ota_tb:1	dB20(VF("/VOUT"))	
1	my_labs:ota_tb:1	ymax(dB20(VF("/VOUT")))	5.71
1	my_labs:ota_tb:1	ymax(mag(VF("/VOUT")))	1.93
1	my_labs:ota_tb:1	bandwidth(VF("/VOUT") 3 "lo...	2.588M
1	my_labs:ota_tb:1	gainBwProd(VF("/VOUT"))	5.007M
Parameters: CIN=12p			
2	my_labs:ota_tb:1	/VOUT	
2	my_labs:ota_tb:1	dB20(VF("/VOUT"))	
2	my_labs:ota_tb:1	ymax(dB20(VF("/VOUT")))	11.43
2	my_labs:ota_tb:1	ymax(mag(VF("/VOUT")))	3.729
2	my_labs:ota_tb:1	bandwidth(VF("/VOUT") 3 "lo...	1.341M
2	my_labs:ota_tb:1	gainBwProd(VF("/VOUT"))	5.012M

Metric	(CIN = 4 pF) $\beta = 0.5$		(CIN = 12 pF) $\beta = 0.25$	
	Simulation	Hand analysis	Simulation	Hand analysis
DC Gain (dB)	1.93 (5.71 dB)	1.929 (5.71 dB)	3.729 (11.43 dB)	3.729 (11.43 dB)
Bandwidth (Hz)	2.588 M	2.59 M	1.314 M	1.34 M
GBW (Hz)	5.007 M	5.006 M	5.0012 M	5.001 M

Larger C_{IN} meaning lower B that will get high gain but lower bandwidth and GBW is fixed if the βAol is high enough we get estimate gain by $1/\beta$

2) Loop Gain vs. Frequency



- hand analysis

$LG = \beta A_{ol}$

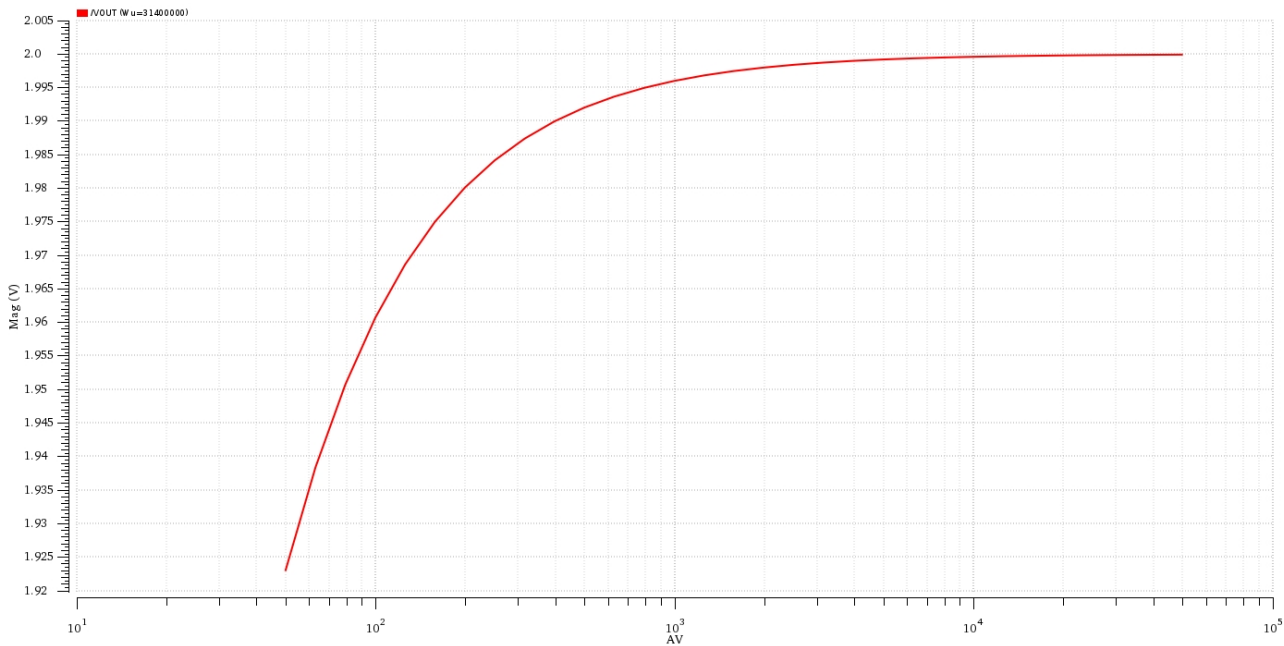
$GBW = f_u = \beta f_u,$

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	my_labs:ota_tb:1	dB20(mag(getData("loopGai...	
1	my_labs:ota_tb:1	ymax(mag(getData("loopGai...	27.5
1	my_labs:ota_tb:1	ymax(dB20(mag(getData("lo...	28.79
1	my_labs:ota_tb:1	unityGainFreq(mag(getData(...	2.5M
Parameters: CIN=12p			
2	my_labs:ota_tb:1	dB20(mag(getData("loopGai...	
2	my_labs:ota_tb:1	ymax(mag(getData("loopGai...	13.75
2	my_labs:ota_tb:1	ymax(dB20(mag(getData("lo...	22.77
2	my_labs:ota_tb:1	unityGainFreq(mag(getData(...	1.249M

Metric	(CIN = 4 pF) $\beta = 0.5$		(CIN = 12 pF) $\beta = 0.25$	
	Simulation	hand analysis	Simulation	Hand analysis
DC Loop Gain (dB)	27.5 (28.79)	27.5 (28.79)	13.75 (22.77)	13.75 (22.77)
GBW (Hz)	2.5 M	2.5 M	1.249 M	1.25 M

Larager CIN meaning lower B that will get a lower DC LG and GBW because proportional to β .

3) Gain Desensitization

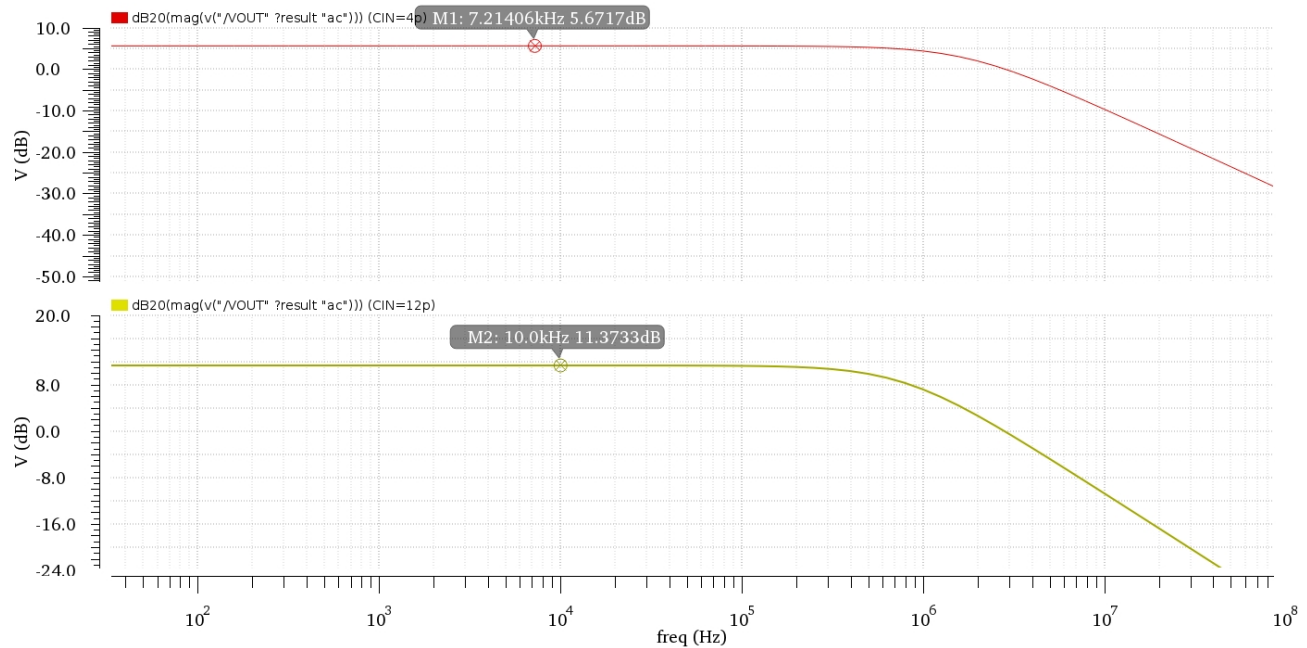


Quantity	Value
Closed-loop gain @ Av = 50 (magnitude)	1.923
Closed-loop gain @ Av = 50000 (magnitude)	2
Ideal closed-loop gain (1 + CIN/CF)	2
% change in closed-loop gain across the Av sweep	4 %

The closed loop gain becomes less sensitive with variation in AOL when it is become high value .
The closed loop gain changed by 4% when the AOL changed from 50 to 50000

Part 2: Feedback with Real 5T OTA

1) Closed-Loop Gain vs. Frequency



- hand analysis

$$Acl = 1 + \frac{CIN}{CF}$$

$$BW = fp (1 + \beta Aol),$$

$$fp = \frac{1}{Rout Cout total}$$

$$cout = CL + C_{feedback}$$

$$C_{feedback} = \frac{CF CIN}{CF + CIN}$$

$$GBW = Acl * BW,$$

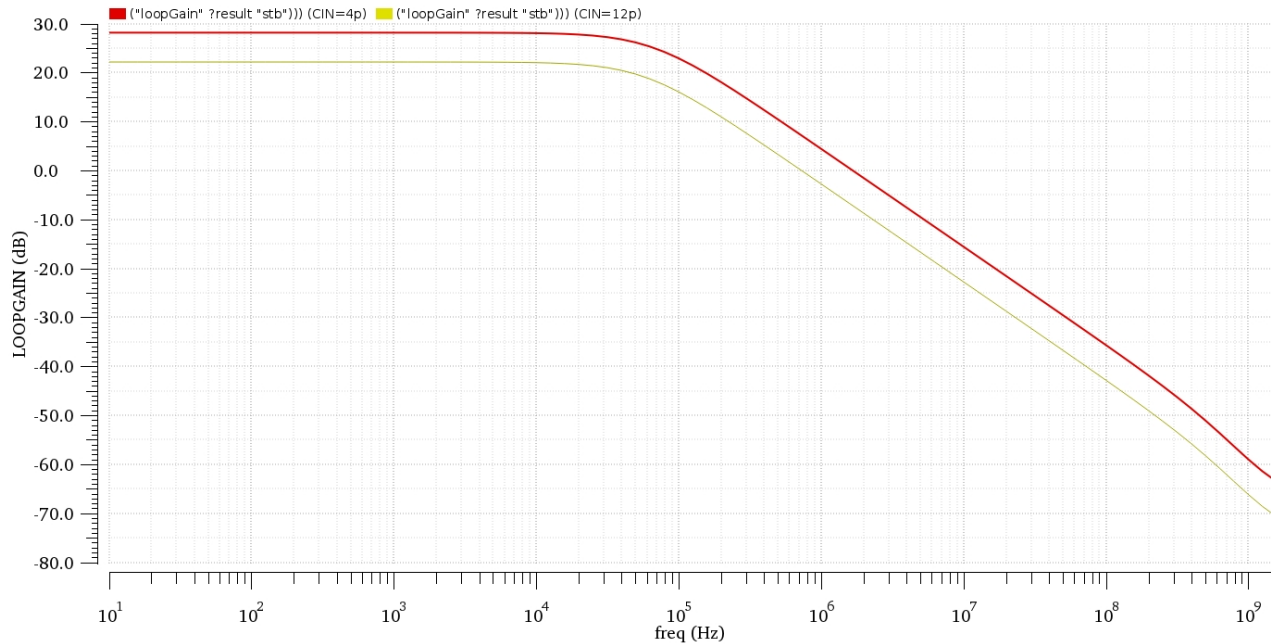
Parameters: CIN=4p		
1	/VOUT	
1	ymax(dB20(VF("/VOUT")))	5.672
1	ymax(mag(VF("/VOUT")))	1.921
1	bandwidth(VF("/VOUT") 3 "low")	1.763M
1	gainBwProd(VF("/VOUT"))	3.395M
Parameters: CIN=12p		
1	/VOUT	
1	ymax(dB20(VF("/VOUT")))	11.37
1	ymax(mag(VF("/VOUT")))	3.704
1	bandwidth(VF("/VOUT") 3 "low")	797.2k
1	gainBwProd(VF("/VOUT"))	2.96M

Metric	(CIN = 4 pF) $\beta = 0.5$		(CIN = 12 pF) $\beta = 0.25$	
	Simulation	Hand analysis	Simulation	Hand analysis
DC Gain (dB)	1.921 (5.672 dB)	2 (6.02 dB)	3.729 (11.43 dB)	4 (12.04 dB)
Bandwidth (Hz)	1.763 M	1.887 M	797.2 k	957.6 k
GBW (Hz)	3.395 M	3.77 M	2.96 M	3.83 M

You will notice that the bandwidth, and consequently the GBW are much smaller than Part 1. Why? the feedback capacitors loads the output of the 5T OTA. That will increase the overall output capacitance then the output pole and bandwidth will decrease. Consequently the GBW will decrease also .

in hand analysis we assume LG is infinity but it is not infinity so there will be error between simulation and it. Larger CIN meaning lower β that get high gain and lower bandwidth

2) Loop Gain vs. Frequency



- hand analysis

$$LG = \beta A_{ol} , A_{ol} = 55$$

$$GBW = \beta f_{u_{OTA}}$$

we will use the f_u for previous closed loop simulation

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	my_labs:ota_tb:1	unityGainFreq(mag(getData(...	1.71M
1	my_labs:ota_tb:1	ymin(mag(getData("loopGai...	25.76
1	my_labs:ota_tb:1	ymin(dB20(mag(getData("lo...	28.22
Parameters: CIN=12p			
2	my_labs:ota_tb:1	unityGainFreq(mag(getData(...	746.8k
2	my_labs:ota_tb:1	ymin(mag(getData("loopGai...	12.93
2	my_labs:ota_tb:1	ymin(dB20(mag(getData("lo...	22.23

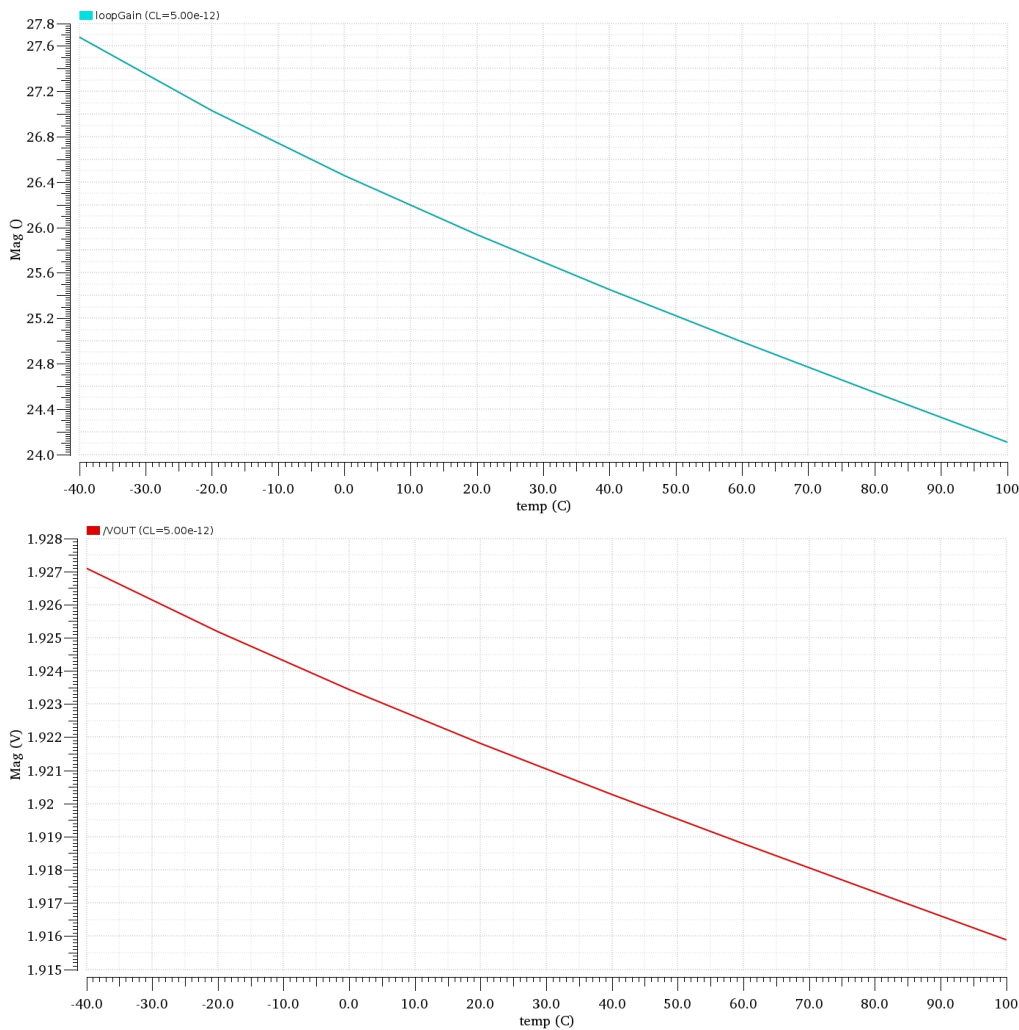
Metric	(CIN = 4 pF) $\beta = 0.5$		(CIN = 12 pF) $\beta = 0.25$	
	Simulation	hand analysis	Simulation	Hand analysis
DC Loop Gain (dB)	25.76 (28.22)	27.5 (28.79)	12.93 (22.77)	13.75 (22.77)
GBW (Hz)	1.7 M	1.885 M	746 k	957 k

Why is the unity-gain frequency much smaller than in Part 1?

the feedback capacitors loads the output of the 5T OTA. That will increase the overall output capacitance so the GBW will decrease also .

Larger CIN meaning lower β that will get a lower DC LG and GBW because proportional to β .
 note also that β is function of frequency so bandwidth will vary also not only dc loop gain

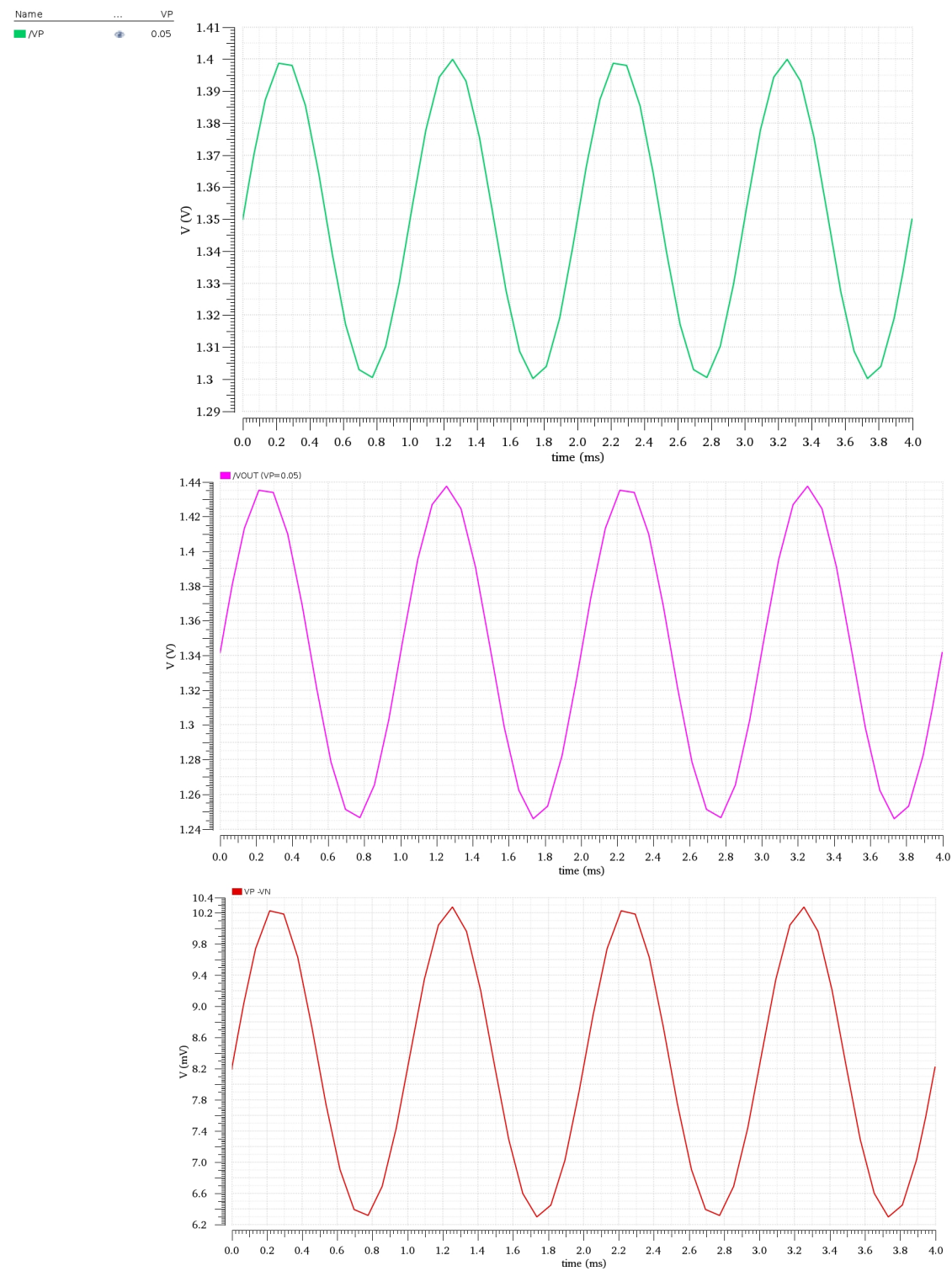
3) Gain Desensitization vs. Temperature



Quantity	Value
% change in DC loop gain	13 %
% change in DC closed-loop gain	0.57 %

We can see the effect of negative feedback that the dc closed loop gain have much less sensitviity to change in temperture .

4) Transient Analysis (vin ,vout , vin_differential)
- fin = 1k



my_labs:ota_tb:1	peakToPeak(v("/VP" ?result "tran"))	99.74m
my_labs:ota_tb:1	peakToPeak(v("/VN" ?result "tran"))	95.75m
my_labs:ota_tb:1	peakToPeak((v("/VP" ?result "tran") - v("/VN" ?result "tran")))	3.984m
my_labs:ota_tb:1	peakToPeak(v("/VOUT" ?result "tran"))	191.5m

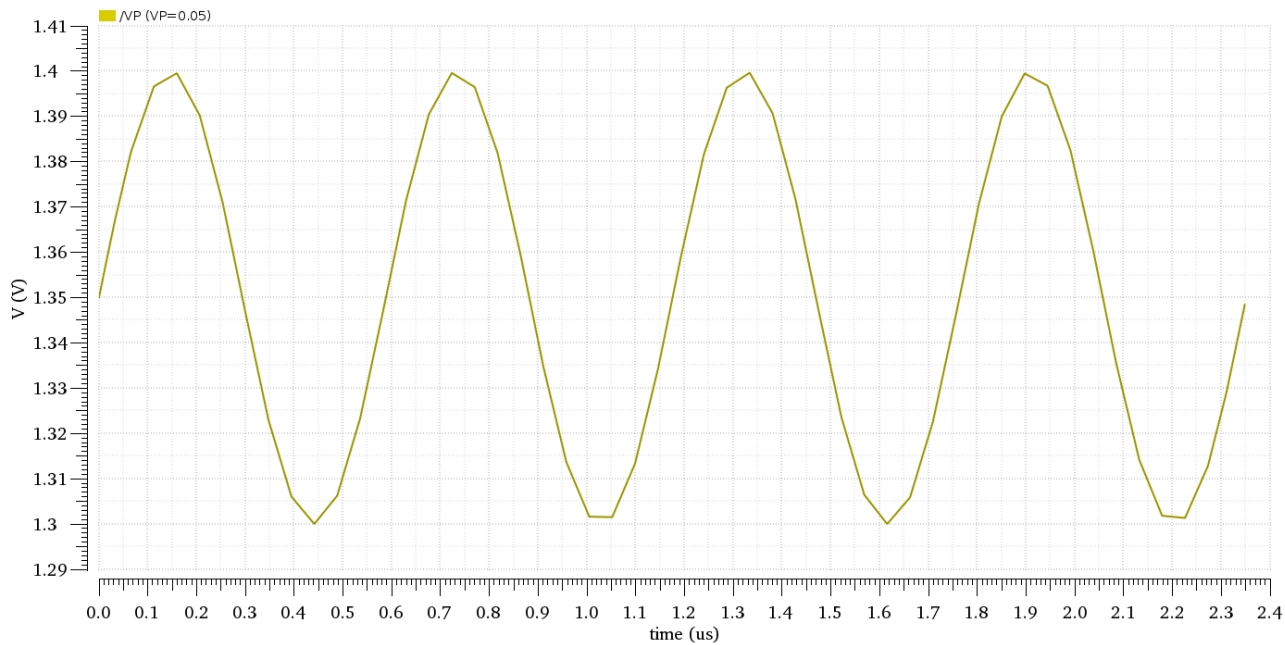
Signal	Peak-to-Peak Voltage
Input (VP , peak to peak)	99.74 m
Output (Vout , peak to peak)	191.5 m
Differential input of OTA (VP – VN , peak to peak)	3.984 m

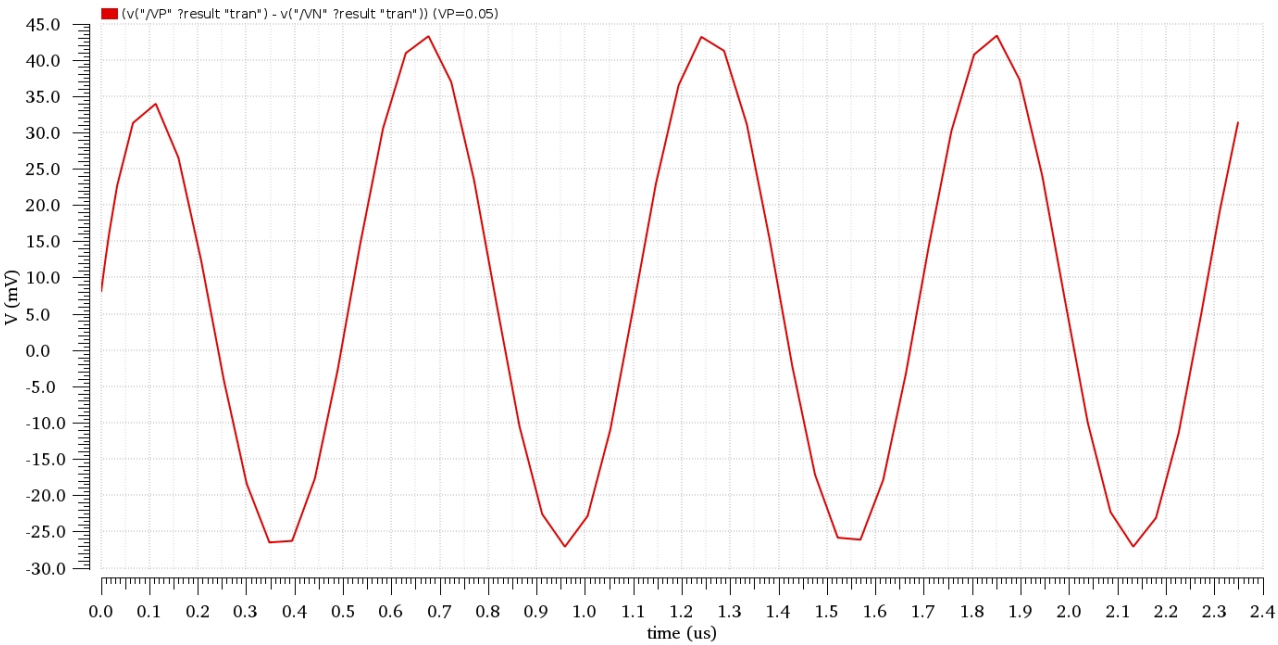
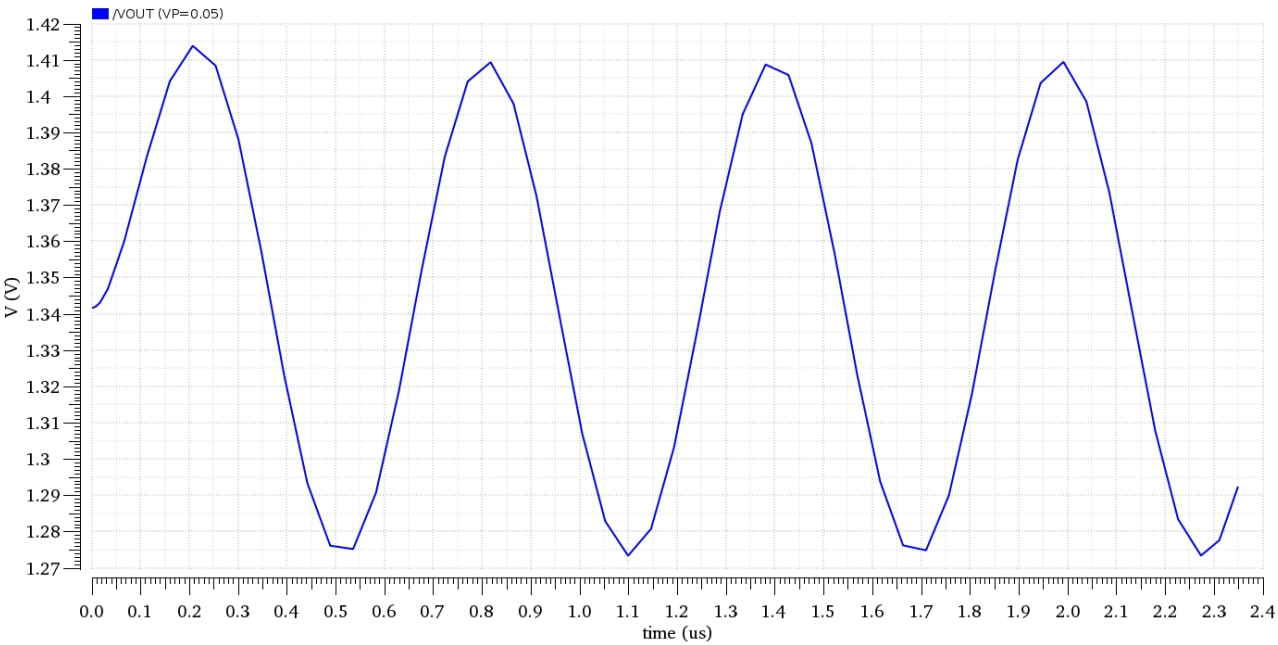
Relation between the output and the differential input (VP – VN)?

$V_{out} / (V_P - V_N) = 48.67$ that aproximatly equal to the dc open gain Aol

$V_{out} / V_P = 1.91$ that equal to dc closed loop gain

- case 2 (fin =closed loop bandwidth)





my_labs:ota_tb:1	peakToPeak(v("/VP" ?result "tran"))	99.65m		
my_labs:ota_tb:1	peakToPeak(v("/VN" ?result "tran"))	70.26m		
my_labs:ota_tb:1	peakToPeak((v("/VP" ?result "tran") - v("/VN" ?result "tran")))	70.44m		
my_labs:ota_tb:1	peakToPeak(v("/VOUT" ?result "tran"))	140.5m		

Signal	Peak-to-Peak Voltage
Input (VP , peak to peak)	99.65 m
Output (Vout , peak to peak)	140.44 m
Differential input of OTA (VP – VN , peak to peak)	29.39 m

Relation between the output and the differential input (VP – VN)?

$V_{out} / (V_P - V_N) = 4.77$, as negative feedback increase the bandwidths so closed loop have higher bandwidth of open loop gain but both of them have the same GBW so they have same asymptotes at high

frequency , which the open loop gain will be as the formula $AOL = Aol / \sqrt{1 + (\frac{f}{f_{ol}})^2}$
